

First-Come,First-Serve CPU Scheduling Algorithm

Completed

Write a C program to implement the First-Come, First-Served (FCFS) CPU Scheduling Algorithm. The program should take the number of processes, their Arrival Times (AT), and Burst Times (BT) as input. Calculate and display the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process, along with the Average Turnaround Time and Average Waiting Time.

To evaluate the performance of the algorithm, we use the following formulas:

- **Completion Time (CT):** The time at which a process finishes execution.
- **Turnaround Time (TAT):** Total time taken from arrival to completion.
- $\$TAT = CT - AT\$$
- **Waiting Time (WT):** The time a process spends waiting in the ready queue.
- $\$WT = TAT - BT\$$

Input Format

1. The first input line contains an integer representing the

Select Language : C (GCC 9.2.0)

```

12     for (int i = 0; i < n; i++) {
13         printf("Enter Arrival Time and Burst Time for process %d: ", i+1);
14         scanf("%d %d", &at[i], &bt[i]);
15     }
16
17     int current_time = 0;
18     for (int i = 0; i < n; i++) {
19         // If the process hasn't arrived yet, the CPU waits until its arrival time.
20         if (current_time < at[i]) {
21             current_time = at[i];
22         }
23
24         // Completion Time = Current Time + Burst Time
25         ct[i] = current_time + bt[i];
26         current_time = ct[i];
27
28         // TAT = CT - AT
29         tat[i] = ct[i] - at[i];
30         // WT = TAT - BT
31         wt[i] = tat[i] - bt[i];
32
33         total_tat += tat[i];
34         total_wt += wt[i];
35     }
36
37     printf("\nAverage Turnaround Time: %.2f", total_tat/n);

```

Test Cases



Run Sample Cases



Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2